GENETIC TEST REPORT

Navy Genomic Laboratory

Patient Details:

Name: Andre Holloway
Date of birth: 20 October 1980
Sex: Male
NHS number: 999 097 0631
Sample type: Peripheral blood, EDTA

Test ordered by:

Name: Dr Whitfield
Organisation: Summit Women's Health
Telephone: 01632 960949

Test carried out by:

Laboratory: Navy Genomic Laboratory
Telephone: 01632 960170
Date received: 28 March 2020
Date reported: 23 April 2020
Authorised by: Dr Helen Ashby

Reason for test: Partner is a carrier of cystic fibrosis.

Your Result	Carrier of cystic fibrosis

ABOUT THE TEST

We tested your CFTR gene. People have two copies of it, one inherited from each parent. Some changes in this gene cause cystic fibrosis. Cystic fibrosis is autosomal recessive. A person needs a changed copy of the gene from each parent to be affected. A person with one changed copy is a carrier and usually has no symptoms.

WHAT THIS RESULT MEANS FOR YOU

This test found one changed copy of the CFTR gene. You are a carrier. You do not have cystic fibrosis, and you are not expected to develop it.

Your partner is also a carrier. A child of two carriers has a one in four chance of having cystic fibrosis, and each pregnancy has the same chance.

A genetic counselor can help you and your partner understand these chances and the choices available to you.

NEXT STEPS

- **Arrange genetic counseling for the couple to review the risk to a child and the options before and during pregnancy.**
- **Make sure the partner's full carrier report is reviewed with the patient, because the risk depends on it.**
- **If the couple is pregnant or planning a pregnancy, discuss prenatal testing, such as chorionic villus sampling or amniocentesis.**

MORE INFORMATION AND SUPPORT

Genetic test results can be hard to take in, and some people find them distressing. To talk through your result, contact the doctor who requested the test, or ring the Clinical Genetics Service on 01632 960171.

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TECHNICAL INFORMATION

Name:	Andre Holloway
Date of birth:	20 October 1980
NHS number:	999 097 0631
Lab number:	FG08-90782
Client Ref #:	FG81-01490
Batch #:	FG78-74890
Sample collected:	23 March 2020
Sample received:	28 March 2020

Keep this page for your records. It is written for health professionals, and you do not need to read it.

Test: Cystic Fibrosis Carrier Test

Gene	Zygosity	Inheritance	HGVS Description	Location	Classification
CFTR	Heterozygous	Autosomal recessive	NM_000492.4:c.3188G>A p.Trp1063Ter	Chr7(GRCh38):g.117611629_117611629	Pathogenic

FULL INTERPRETATION

According to the referral, Andre Holloway's partner Lena, born 05/07/1990, is a known carrier of cystic fibrosis, with the CFTR variant c.3728T>A in one copy.

The patient carries one pathogenic change in the CFTR gene. Because there is only one changed copy, the patient is a carrier and is not expected to have cystic fibrosis.

The test was ordered because the partner is a carrier of cystic fibrosis. If both partners carry a changed copy of the CFTR gene, each pregnancy has a one in four chance of a child with cystic fibrosis. The partner's own results are needed to judge this risk, and they should be reviewed with the patient.

TEST METHODOLOGY

The coding exons and flanking intronic sequence of CFTR were sequenced on a next-generation sequencing platform, with reads aligned to the GRCh38 reference genome. HGVS recommendations are followed for variant names, with NM_000492.4 as the reference sequence.

False positive and false negative rates are each about 0.1%.

Reported by Dr Samuel Grey. Authorised by Dr Helen Ashby.